REHAN AHMED

Karachi, Pakistan · Email · LinkedIn · GitHub

Achievements & Awards

Mitacs Globalink Research Award, Mitacs Canada 2026

- Competitive international award for a research placement at the University of Lethbridge, Alberta (Supervisor: Dr. Sidney Shapiro) on interpretable machine learning for assessing the commercialization readiness of early-stage deep-tech ventures.

Vice Chancellor's Honour List (2×), DHA Suffa University — 4.0 semester GPA Fall 2024, Spring 2025

Dean's Honour List, DHA Suffa University — 3.9 semester GPA, ranked top of class Spring 2024

IELTS Academic: Overall Band 7.0 (CEFR C1) 2026

Education

DHA Suffa University Karachi, Pakistan

BS in Artificial Intelligence, CGPA: 3.82 / 4.0 *Expected July 2027*

Research Experience

Turing Science Labs Remote

Undergraduate Researcher — Scaling Laws for Vision-Language Models under Label Noise *Aug 2026 – Present*

Advisor: Dr. Vinoth Nandakumar

- Investigating how vision-language models trained from scratch behave as model size, label noise, and data complexity vary.
- Building synthetic Urdu visual question-answering datasets to run these experiments in a controlled, low-resource language setting.
- Manuscript in preparation for Transactions on Machine Learning Research (TMLR).

DHA Suffa University — SUPARCO Collaboration Karachi, Pakistan

Research Assistant, Image Enhancement Models *Mar 2025 – Jul 2025*

- Built and benchmarked SRGAN and SwinIR super-resolution models for $4\times$ upscaling of low-resolution satellite imagery with Pakistan's national space agency.
- Set up reproducible training and evaluation pipelines for comparing models on remote-sensing data.

r2.ii Remote

Research Intern, AI for Health *Feb 2023 – Aug 2023*

- Reproduced and extended published early-sepsis prediction methods on irregularly sampled critical-care time series (MIMIC-III, MIMIC-IV).
- Evaluated competing diagnostic approaches and maintained technical documentation for the team.

Publications & Submissions

[1] U. Mansoor, **R. Ahmed**, and M. Sabih, "When Does Self-Supervised Pretraining Help? Label Efficiency of Joint-Embedding Predictive Representations (JEPA) for Early Sepsis Prediction." *Abstract submitted to BioBIS 2027: Advances in Bioengineering and Biomedical Intelligent Systems Conference*, Saudi Arabia.

[2] U. bin Mansoor, S. M. Rashid, S. R. Naqvi, and **R. Ahmed**, "A Survey of Early Detection and Prediction of Sepsis: From Clinical Scoring Systems to Deep Learning and Real-World Artificial Intelligence." *Submitted to Machine Learning in Medicine*, 2026. (Authors 2–4 contributed equally.) [PDF]

Projects

LegalFlow — *Legal Case-Management Platform with an AI Assistant* Team project

- Built a retrieval-augmented document-understanding pipeline (Gemini embeddings, pgvector, OCR) that searches and summarizes case files and extracts deadlines, parties, and key facts with confidence scores and source links.
- Built LOIS, a tool-calling LLM assistant with 20+ role-scoped tools that executes multi-step requests and drafts letters from case data (Next.js, FastAPI, PostgreSQL).

Ilaaj AI — *End-to-End AI Healthcare Platform* ilaaj-ai.vercel.app

- Trained a chest X-ray classifier on CheXpert achieving 0.89 AUC across five pathologies, deployed in a full-stack diagnostic platform (PyTorch, FastAPI, Streamlit).
- Added OCR-based prescription digitization and automated PDF report generation for patients and clinicians.

More projects: hackathon builds at github.com/Rehanabbaxi/Hackathon-Projects

Technical Skills

Languages:	Python, TypeScript
Frameworks:	PyTorch, TensorFlow, Hugging Face, FastAPI, Streamlit
Methods:	Deep Learning, Learning with Noisy Labels, Scaling Laws, Vision Transformers, GANs, Computer Vision, NLP, RAG and LLM Tool-Calling, Interpretable ML, Medical Imaging
Tools:	Git, Linux, PostgreSQL/Supabase (pgvector), Playwright

Technical Writing & Certifications

Medium — *Technical Articles* medium.com/@rehan020345

- "Understanding Temporal Convolutional Networks (TCNs): From CNN Basics to Full Sequence Mastery"
- "Demystifying SRGAN: Deep Dive into Super-Resolution GANs"

DeepLearning.AI — *Coursera Certifications*

- Mathematics for Machine Learning and Data Science
- AI for Medical Diagnosis
- AI for Medical Prognosis